



DEPARTMENT OF
COMPUTER SCIENCE



FUNCTIONS AND MODULES

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LEARNING OBJECTIVES

At the end of the lesson, you should be able to:

- Discuss functions and function properties.
- Describe unions and relate them to functions.
- Apply function and module concepts to problem solving.



FUNCTIONS

Having to solve large and complex computing projects means that it will be either impossible or time wasting for only one person to deliver a solution quickly.

The more complex the computing task, the more people will be required to work on it.



FUNCTIONS

So, how do we ensure that lots of people work on a single project without issues? Break down the complex computing projects into smaller ones.

Functions are used to break large and complex computing projects into smaller ones such that different people can work on smaller functions at the same time.



FUNCTIONS

A function is a group of statements that perform a task.

Programme codes can be separated into functions.

How the code is separated among different functions is up to the programmer, but logically the division is such that each function performs a specific task.



FUNCTION DECLARATION

A function declaration tells the compiler about a function name, return type, and parameters. The actual body of the function can be defined separately.

```
return_type function_name( parameter list );
```

```
int max(int num1, int num2);
```



FUNCTION DEFINITION

A function definition provides the actual body of the function.

```
return_type function_name( parameter list ) {  
    |   body of the function  
}
```

A function definition consists of a function header and a function body. Here are all the parts of a function

- Return Type: The data type of the value the function returns. The return_type can also be void.



FUNCTION DEFINITION

```
return_type function_name( parameter list ) {  
    |   body of the function  
}
```

- Function Name: This is the actual name of the function.
- Parameters: A parameter is like a placeholder. When a function is invoked, a value is passed to the parameter. A function may contain no parameters.



FUNCTION DEFINITION

```
return_type function_name( parameter list ) {  
    |   body of the function  
}
```

- **Function Body:** The function body contains a collection of statements that define what the function does.



CALLING A FUNCTION

Calling a function means that we want to use the function. What does this mean?

```
int max(int num1, int num2);
```

```
/* Variable definition */  
int a = 100;  
int b = 200;  
  
/* calling a function to get max value */  
int maxNumber = max(a, b);
```



BENEFITS OF USING FUNCTIONS

- Functions allow us to scale up a project easily because even if a small project rapidly becomes complex, we would only need to add more functions and the system will remain stable.
- Functions allow us to avoid duplicating codes, because regardless of how many times a function is used in a project, we only need to implement it once.



MODULES

Modules and functions are very similar. The main difference is that modules are used on a much larger scale.

While functions are used specifically to solve specific problems on a smaller scale, modules are used on a larger scale because they can contain various functions and classes.



MODULES

To use modules in our code, we need to import the module. For instance, let us assume the 'max' function we implemented earlier is part of a module file called 'special_functions'. To use any function in the 'special_functions' module in our code we need to import it by using commands such as 'import special_functions'.

```
import special_functions;
```



MODULES

If we want to explicitly specify the exact function we want to use in the 'special_functions', we can use commands such as 'from special_functions import max'. With this command, we can only use the 'max' function in the 'special_functions' because we explicitly specified it.



SUMMARY

- A function declaration tells the compiler about a function name, return type, and parameters.
- A function is a group of statements that perform a task.
- Functions allow us to avoid duplicating codes, because regardless of how many times a function is used in a project, we only need to implement it once.



FURTHER READING RESOURCES

- Kernighan, B. W., & Ritchie, D. M. (2002). The C programming language.
- Bailey, T. (2005). An introduction to the c programming language and software design. July-2005



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THANK
YOU